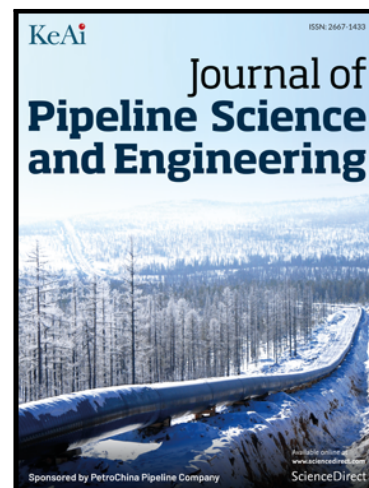


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Risk Transmission Mechanism and Prevention Framework of Hydrogen Pipeline: A Dual-Drive Hybrid Approach

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Abstract: Hydrogen energy stands as a pivotal secondary energy carrier for achieving carbon neutrality, yet its high propensity for leakage, ignition, and cascading explosions presents unprecedented multi-state, low-probability-high-consequence risks for pipeline transportation. Current research lacks quantitative characterization of chain-reaction failure mechanisms in hydrogen pipelines, impeding precision risk mitigation. Based on an analysis of 50 global hydrogen pipeline accident cases, this study applies the Delphi method to identify and extract 45 high-frequency risk events that critically influence hydrogen infrastructure failures. We propose a dual-driven hybrid model that integrates complex network theory with association rule mining. Using complex network theory, we constructed the topological structure of the initial disaster chain network and identified the top 10 critical nodes through composite centrality indices. A Frequent Pattern Growth (FP-Growth) algorithm based on risk importance was implemented to investigate coupling relationships between risk sources and accident consequences, revealing intrinsic accident mechanisms. By mapping these association rules onto a directed weighted network, we identified two high-risk disaster chains. Critical findings show that gasket fatigue and seal failure are primary initiators of leakage, while the transition from leakage to explosion represents a core escalation pathway. By fusing global network metrics with localized FP-Growth rules, our approach quantitatively locates silent risks in hydrogen pipelines. The proposed three-tier prevention system encompasses source control, process interruption, and terminal emergency response. It is designed for seamless integration within integrity management cycles. This research provides both a methodological framework and decision-support tools for scalable risk governance of hydrogen infrastructure, addressing a critical gap in sustainable energy transition safety protocols.

Keywords: Hydrogen gas pipeline; Disaster chain system; Complex network; FP-Growth algorithm;

Disaster prevention strategies

1. Introduction

Amidst the escalating global energy demand and increasingly severe environmental challenges, the development of clean energy has become a strategic priority for nations worldwide Dehdari, L., Burgers, I., Xiao, P., Li, K. G., Singh, R., Webley, P. A., 2022. Purification of hydrogen from natural gas/hydrogen pipeline mixtures. *Separation and Purification Technology*. 282, 120094. Russo, P., De Marco, A., Parisi, F., 2020. Assessment of the Damage from Hydrogen Pipeline Explosions on People and Buildings. *Energies*. 13, 5051. . In this context, hydrogen energy, owing to its zero-carbon emissions, renewability, and superior energy density, is emerging as a pivotal component of future energy systems (Witkowski, A., Rusin, A., Majkut, M., Stolecka, K., 2017. Comprehensive analysis of hydrogen compression and pipeline transportation from thermodynamics and safety aspects. *Energy*. 141, 2508-25. Compared to conventional fossil fuels, hydrogen combustion offers a higher heat value while producing only water vapor as a byproduct, effectively addressing greenhouse gas emissions (Viana, F. F. C. L., Alencar, M. H., Ferreira, R. J. P., De Almeida, A. T., 2022. Multidimensional risk assessment and categorization of hydrogen pipelines. *International Journal of Hydrogen Energy*. 47(42), 18424-18440.; Ueckerdt, F., Verpoort, P. C., Anantharaman, R., Bauer, C., Beck, F., Longden, T., Roussanly, S., 2024. On the cost competitiveness of blue and green hydrogen. *Joule*. 8(1), 104-128. Li, Y., Kuang, Z., Fan, Z., Shuai, J., 2023. Evaluation of the safe separation distances of hydrogen-blended natural gas pipelines in a jet fire scenario. *International Journal of Hydrogen Energy*. 48(49), 18804-18815. Zhang, X., Yang, Z., Huang, X., Wang, X., Pan, Y., Zhou, X., 2021. Combustion enhancement and inhibition of hydrogen-doped methane flame by HFC-227ea. *International Journal of Hydrogen Energy*. 46(41), 21704-21714..

Pipeline transportation represents one of the most efficient methods for hydrogen delivery, offering significant cost advantages over liquid hydrogen transport and compressed hydrogen gas transport. The US and Europe currently dominate global hydrogen pipeline infrastructure, operating approximately 5000 km of dedicated lines, with 2700 km across the US and 1600 km throughout Europe (Cao, Q., Wang, H., Qin, Y., Wang, M., 2024. Current status and analysis of the development of pure hydrogen pipeline hydrogen transmission technology. *Mechanics in Engineering*. 46(1), 18-

27. . By 2040, there is a plan to have 40,000 km of hydrogen pipelines in Europe (Dao et al., 2024). China has also achieved remarkable progress in both pure hydrogen pipeline construction and hydrogen-blended natural gas pipelines, driving rapid industrial expansion. By the end of 2024, 30 provincial-level administrative regions in mainland China have incorporated hydrogen energy into their 14th Five-Year Plans, with many issuing specialized hydrogen industry policies. In April 2023, China National Petroleum Corporation launched a hydrogen-natural gas blending demonstration project in Ningdong, Yinchuan, achieving a hydrogen volume fraction of up to 24% (Yang, M., Li, L. D., Zhang, W. K., Lin, X. F., Cao, Q., Zhang, X. R., 2025. Development status and analysis of hydrogen pipeline transportation technology. *Modern Chemical Industry*. 45(5), 25-30, 36. . By November 2024, China first high-pressure, long-distance hydrogen-blended natural gas pipeline, the Baotou-Linhe Pipeline in Inner Mongolia, has become fully operational (Editorial Board., 2024. China's first high-pressure pipeline with hydrogen blending was ignited. *Chemical Engineering Management*. (35), 93.. In January 2025, with the approval of the Development and Reform Commission of the Xinjiang Uygur Autonomous Region, the National Pipeline Network Group West Pipeline Company established the Xinjiang Hydrogen Storage and Transportation Engineering Research Center. This center provides a key platform for research and development in hydrogen pipeline technology.

However, hydrogen storage and transportation still face significant safety challenges. Compared to methane, hydrogen has a smaller molecular size and lower viscosity and density. These properties increase its risk of leakage through pipelines and storage materials, as well as the potential for seal failure. Furthermore, these specific molecular characteristics can promote hydrogen-induced cracking in metals and other materials, which may lead to catastrophic failures (El-Kady et al., 2024). Hydrogen exhibits distinctive physicochemical traits that markedly intensify potential hazards (Nykyforchyn, H., Zvirko, O., Hredil, M., Krechkovska, H., Tsyrlunyk, O., Student, O., Unigovskyi, L., 2022. Methodology of hydrogen embrittlement study of long-term operated natural gas distribution pipeline steels caused by hydrogen transport. *Fracture and Structural Integrity*. 16(59), 396-404.. With a minimum ignition energy of just 0.017 mJ and flammability limits extending from 4% to 75% , hydrogen exhibits exceptional combustion reactivity (Ruiz-Tagle, A., Groth, K. M., 2024. Comparing the risk of third-party excavation damage between natural gas and hydrogen pipelines. *International Journal of Hydrogen Energy*. 57, 107-120., leaked hydrogen can be easily ignited even by weak ignition sources, resulting in a high probability of combustion. Once released, hydrogen poses a

substantial risk of fire or explosion, potentially leading to catastrophic casualties and economic losses. A thorough investigation into the evolution mechanisms of hydrogen pipeline failures is therefore essential. This study aims to identify and quantify critical risk factors, with particular emphasis on those characterized as low-probability yet high-consequence "silent risk points". It also provides a foundation for establishing a scientific and reliable system for hazard prevention and control.

Disasters typically manifest as chain-like or interconnected network structures. As a crucial methodology for investigating complex disaster systems, disaster chain theory has been widely applied to analyze the evolution processes of both natural and anthropogenic hazards. The predominant analytical approaches include complex network models, probabilistic models, and physical disaster chain models. In recent years, researchers have begun applying disaster chain theory to pipeline hazard analysis. Jin, Y., Shan, K., Li, Y., Shuai, J., 2024. Analysis of urban gas pipeline leakage disaster evolution process based on disaster chain and complex network theory. *Safety and Environmental Engineering*. 31(2), 34-43. integrated disaster chain theory with complex network analysis to identify and quantify critical nodes and edges in urban gas pipeline leakage disaster chains, enabling systematic risk assessment. Wang, C., Wang, Z., 2022. Risk analysis of oil pipeline leakage disaster chain at the wildland-urban interface. *Industrial Minerals & Processing*. 51(4):19-25. developed a complex network model for oil pipeline leakage disaster chains, conducting comprehensive risk analysis at macro-levels, meso-levels, and micro-levels. Wang, C., Lv, S., 2016. Research on disaster chain of leakage disaster in urban gas pipeline. *Journal of Safety Science and Technology*. 12(5):16-21. employed Markov chain theory to investigate the disaster formation process and chain mechanisms in gas pipeline leaks, quantitatively assessing failure probabilities. Feng, Y., Gao, J., Yin, X., Chen, J., Wu, X., 2024. Risk assessment and simulation of gas pipeline leakage based on Markov chain theory. *Journal of Loss Prevention in the Process Industries*. 91, 105370. constructed an evolutionary disaster chain for urban gas network leaks, utilizing Bayesian network modeling to analyze the disaster progression mechanisms. Liu, H., 2024. Research on the probability assessment method of earthquake secondary fire based on disaster chain principle. Institute of Engineering Mechanics, China Earthquake Administration. established physical disaster chain models for civil gas pipeline fires, examining seismic-induced secondary disaster mechanisms while evaluating occurrence probabilities through Bayesian networks. Xu, Z. D., Liu, X., Xu, W., Sun, B., Liu, X., Xu, D., 2022. Analysis on the disaster chain evolution from gas leak to explosion in urban utility tunnels. *Engineering Failure Analysis*. 140,

106609. proposed a physics-based disaster chain model characterizing the complete evolution process from methane leakage to explosion in utility tunnels. Despite their value for general pipeline disaster chains, research targeting the unique characteristics of hydrogen pipeline systems is still in a nascent state, constituting a significant knowledge gap.

It is worth noting that traditional disaster chain analysis mostly relies on qualitative descriptions or simplified quantitative models. When dealing with highly complex and multi-factorally coupled disaster networks, it is often difficult to effectively uncover the underlying deep correlations and evolution patterns hidden within them. With the rapid development of artificial intelligence technologies, intelligent identification and hazard management have emerged as crucial research directions in industrial safety (Liu, W.; Luo, R.; Xiao, M.; Chen, Y. Intelligent detection of hidden distresses in asphalt pavement based on GPR and deep learning algorithm. *Constr. Build. Mater.* 2024, 416, 135089.). Advanced algorithms for hazard data mining and analysis can significantly enhance the accuracy and efficiency of risk identification. Among machine learning approaches, association rule mining algorithms have demonstrated exceptional capability in discovering relationships among different items or attributes (Yu, L., 2018. Preliminary study on application of data mining technology in analysis of institutional investors' investment behavior. *Science & Technology Information.* 16(32), 22-23. . Such algorithms can automatically identify frequent item sets and strong association rules from massive data. This capability makes them a powerful tool for quantitatively analyzing the complex causal relationships within multi-node, multi-path disaster chains. These algorithms have been successfully applied across various domains including logistics, financial systems (Yang, B., 2020. Construction of logistics financial security risk ontology model based on risk association and machine learning. *Safety Science.* 123, 104437., intelligent recommendation systems (Mulyawan, B., Vionelsy., Sutrisno, T., 2020. Product recommendation system on building materials shopping using FP-Growth algorithm. In *Proceedings of the 2020 International Conference on Materials Science.* 7, 012144. , and healthcare (Xu, F., Lu, H., 2017. The application of FP-growth algorithm based on distributed intelligence in wisdom medical treatment. *International Journal of Pattern Recognition and Artificial Intelligence.* 31(4), 1759005. ; Guo, H., Liu, H., Chen, J., Zeng, Y., 2021. Data mining and risk prediction based on apriori improved algorithm for lung cancer. *Journal of Signal Processing Systems.* 93(7), 795-809.. Their potential in safety risk detection and control presents a viable solution to the complex challenges posed by hydrogen pipeline disaster chain analysis.

In the field of industrial safety, researchers have made innovative applications of association rule mining algorithms. Li, S., You, M., Li, D. Liu, J., 2022. Identifying coal mine safety production risk factors by employing text mining and Bayesian network techniques. *Process Saf. Environ. Prot.* 162, 1067-1081. developed a Bayesian network model incorporating text mining and strong association rules, conducting sensitivity analysis and critical path analysis to identify root causes of mining accidents. Jing, G., Qin, H., Jiang, F., 2023. Coal mine safety accident analysis based on Apriori algorithm. *Journal of Safety and Environment.* 24(6), 2313-2320. employed Apriori algorithm with Gephi visualization to statistically analyze Chinese coal mine accident reports , revealing complex inter-factor relationships. Guo, A., Yang, Z., Du, K., Zhang, X., 2023. Design and application of Coal mine safety information management system based on FP-Growth algorithm. *China Mining Magazine.* 32(3), 67-73. applied FP-Growth algorithm for in-depth mining of accident precursor data from Shaanbei coal mines, demonstrating its effectiveness in early warning and risk prevention. Tao, F., Pi, Y., Zhang, M., Yuan, C., Deng, M., 2023. Hidden Danger Association Mining for Water Conservancy Projects Based on Task Scenario-Driven. *Water*, 15(15), 2814. modified the traditional Apriori algorithm to develop a task-scenario-based association mining method for hydraulic engineering hazards, successfully implemented in the Xinmeng River dredging project. Cai, J., Song, R., He, S., Zhao, R., Jiang, J., 2024. Construction safety risk analysis of business lines based on improved FP-Growth algorithm and Bayesian. *Journal of Railway Science and Engineering.* 21(8), 3370-3381. modified the FP-Growth algorithm using compression techniques and differential encoding to identify high-lift, high-support, and high-confidence rules in railway construction accidents. Shi, G., Ding, X., Hong, C., Liu, Z., Zhao, L., 2024. Research on key risk chain mining method for urban rail transit operations: A new approach to risk management. *International Journal of Transportation Science and Technology.* 13, 29-43. proposed an modified H-Apriori algorithm incorporating disaster severity weighting, enabling effective discovery of strong association rules among risk sources for urban rail transit management. Wang, C., Fu, A., Li, W., Li, M., Chen, T., 2024. Intelligent Identification of Hidden Dangers in Hydrogen Pipeline Transmission Station Using GWO-Optimized Apriori Algorithm. *Energies.* 17(18), 4539 developed a Grey Wolf Optimizer-enhanced Apriori algorithm for mining association rules in hydrogen pipeline transmission station hazards.

Numerous scholars have conducted extensive foundational research on risk chain analysis and critical hazard source identification in engineering fields. However, significant gaps persist. Current

studies on risk disaster chains in hydrogen pipelines remain scarce. The current body of research often remains at the level of summarizing risk correlations and qualitatively assessing methodologies or network topologies, rather than advancing toward data-driven excavation of risk chains from the intrinsic features of the data itself.

The frequency-counting nature of traditional association rule algorithms poses a significant obstacle. It impedes their ability to accurately identify strong association rules between pipeline risk precursors and accidents, despite their widespread use. Computationally, Apriori-type algorithms require multiple database scans and generate excessive candidate itemsets, leading to combinatorial explosion when analyzing long transaction sequences. Regarding engineering applicability, FP-Growth exhibits sensitivity to sparse data and struggles to process continuous risk parameters directly.

Hydrogen pipelines present a unique multi-state risk profile, including mechanical damage, hydrogen embrittlement, and third-party interference. We propose an integrated risk mining framework. This framework achieves breakthroughs through three key innovations. First, it clearly differentiates the coupling relationships between heterogeneous risk sources. Second, it introduces a risk-weighted mechanism. This mechanism balances the conflict between consequence severity and occurrence frequency. Third, it embeds a dynamic pruning strategy into the FP-Tree structure. This strategy effectively controls memory usage when mining high-dimensional and heterogeneous accident data. Our solution not only can accurately extract the key risk chains, but also can quantify their aggregated risks, laying a solid foundation for subsequent prevention and mitigation strategies.

The paper is structured as follows. Section 1 briefly reviews the state of knowledge on pipeline disaster chains and risk mining. Section 2 formulates a four-dimensional coupled disaster chain tailored to hydrogen pipelines. Section 3 presents the risk-mining model that extracts and quantifies critical chains embedded in the disaster network. Section 4 applies the framework to some cases of pipeline accidents, combining complex-network centrality metrics with association-rule mining to identify pivotal nodes and dominant chains. Section 5 translates these findings into optimized prevention and mitigation recommendations. Finally, Section 6 concludes the study and outlines avenues for future research.

2. Disaster Chain System Theory Model

Disaster chain theory provides a critical framework for understanding the evolution of complex disasters. It conceptualizes catastrophes not as isolated events, but as dynamic systems of intricate

feedback loops. These loops interconnect internal system components, link different disaster systems, and couple them with external environmental conditions throughout their entire lifecycle, from gestation to regression (Li, W., Jin, S., 2021. A study on forecasting the demands of crisis information based on disaster chain theories. *Information Science*. 39(9), 25-34. . This chained interaction profoundly reveals the complex nature of disasters and establishes a theoretical foundation for clarifying the intrinsic chain-propagation relationships among multiple disaster types.

Within hydrogen pipeline systems, the formation and progression of disaster chains are profoundly influenced by their specific operating context. The hazard-forming environment refers to the natural and socio-technical surroundings in which pipelines are situated, serving as the breeding ground for latent risk factors (Wang, C., Wang, Z., 2022. Risk analysis of oil pipeline leakage disaster chain at the wildland-urban interface. *Industrial Minerals & Processing*. 51(4):19-25. . The activity level of the environment directly dictates how rapidly and intensely disaster inducing factors such as material defects, operational errors, or third party damage accumulate. When these factors reach a critical threshold within a specific hazard-forming environment, an initial disaster event may be triggered.

Hydrogen pipeline incidents exhibit unique hazards due to their high propensity for chain-reaction escalation. Owing to hydrogen extremely low explosion limit and minimal ignition energy (Ruiz-Tagle, A., Groth, K. M., 2024. Comparing the risk of third-party excavation damage between natural gas and hydrogen pipelines. *International Journal of Hydrogen Energy*. 57, 107-120., a leakage event can rapidly achieve an explosive concentration in air. Subsequent ignition by minuscule sources can readily initiate fires or violent explosions. Such fires or explosions, directly precipitated by the initial leakage event, constitute classic secondary disasters.

In disaster chain theory, the disaster-bearing carrier is the entity directly subjected to the disaster energy and represents a crucial node determining whether the chain effect is amplified or mitigated. When the intensity of the initial or secondary disaster event exceeds the resilience threshold of the carrier, disaster consequences manifest and may escalate further through chain reactions. During this evolution, disaster energy continuously accumulates, transfers, and releases, leading to intensified destructive power, expanded impact scope, and ultimately more severe disaster consequences (Mao, H., Zhang, L., Luo, P., 2019. Construction of rainfall flood disaster adaptation space in mountainous cities based on disaster chain theory: A case study of urban design of Zaoyang New District in Wushan

County. Landscape Architecture. 26(7), 96-100. .

In **Error! Reference source not found.**, we establishes a four dimensional coupling mechanism termed hazard forming environment, disaster inducing factors, disaster bearing carrier, and disaster consequences to address the inherent susceptibility of hydrogen pipelines to hydrogen embrittlement, leakage, deflagration, and detonation. This mechanism delineates the complete evolution logic of hydrogen pipeline disaster chains.

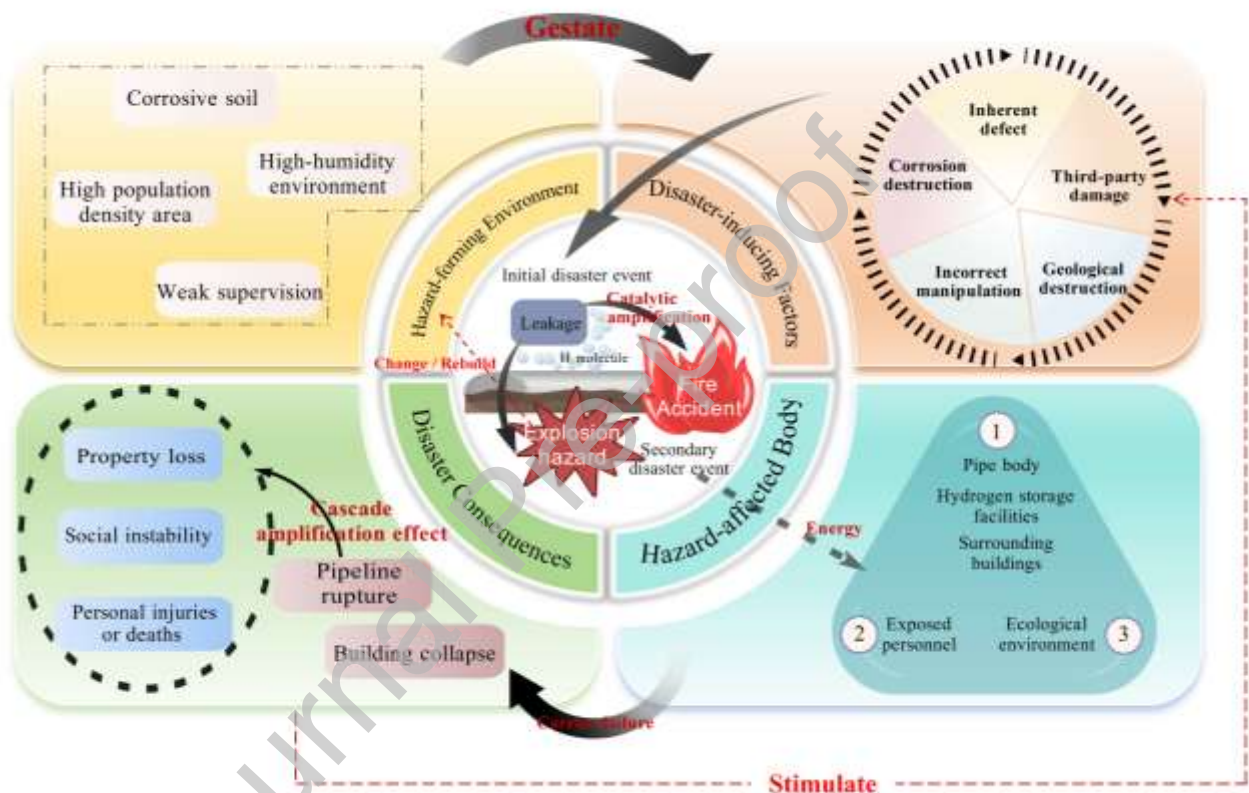


Fig. 1 Mechanism of disaster chain formation of hydrogen pipelines

First, the forward-driving path forms the main sequence of disaster evolution. It describes the progressive transfer of disaster energy and mass. This path typically originates from a specific hazard-forming environment, which fosters the emergence and accumulation of hazard-inducing factors. When these factors exceed the system's resilience threshold, an initial disaster event is triggered. At this stage, the unique physicochemical properties of hydrogen act as a powerful catalytic amplifier. They significantly increase the likelihood and speed of secondary disasters. The energy from both initial and secondary disasters then acts upon disaster carriers. Once the intensity of this action surpasses the carriers' load-bearing capacity, failure occurs. The released disaster energy further affects

more vulnerable entities. Through such cascading amplification effects, severe consequences eventually arise. These include casualties, economic losses, social disruption, and environmental damage.

In reality, disaster evolution is not a one-way process. It is filled with feedback loops that dynamically escalate the disaster. The disaster consequences can intensely feed back, instantly altering or even completely reshaping the original hazard-forming environment. This drastically accelerates disaster escalation and may induce derivative disasters. Simultaneously, the response state of disaster carriers can also feed back. This may stimulate entirely new hazard-inducing factors, creating fresh risk sources. As a result, the disaster chain continues to spread in both time and space. Furthermore, cross-dimensional interactions within the system cannot be overlooked. These interactions cause coupling relationships to exhibit networked characteristics. For example, the hazard-forming environment can directly weaken the inherent resilience of disaster carriers. This directly reduces the structural stability of the pipeline.

3. Disaster Chain Risk Mining Model

3.1 Overall excavation framework

After elucidating the transmission mechanism of hydrogen gas pipeline disaster chains, it is still crucial to explore an effective way to quantify the chain transmission strength to provide a theoretical basis for disaster prevention and mitigation. Based on this, this study innovatively proposes a hybrid analysis framework, as shown in the Fig. 2. Through the integration of complex network theory and the modified FP-Growth algorithm, it systematically solves the difficult problem of identifying the risk of hydrogen gas pipeline disaster chains.

The framework first constructs a data-driven layer. Leveraging a global historical failure case library, we generated a time-serialized transaction database to structurally store the spatiotemporal evolution of accident chains and systematically encode risk source tags. In the algorithm core layer, dual-engine collaborative analysis is performed. On the one hand, based on the system theory of disaster chains, a directed and weighted network is constructed. Hub nodes are quantified using centrality measures in the complex network. In parallel, the modified FP-Growth algorithm integrates a Weighted Support index and dynamic pruning, effectively uncovering high-frequency patterns as well as low-probability, high-consequence risk rules. The disaster chain extraction layer integrates network topology parameters and association rule metrics to generate a key disaster chain graph

representing multidimensional rule features, outputting high-risk sequences and their propagation strength rankings. Finally, in the strategy generation layer, a three-tier prevention and control system is constructed based on the disaster chain disruption principle.

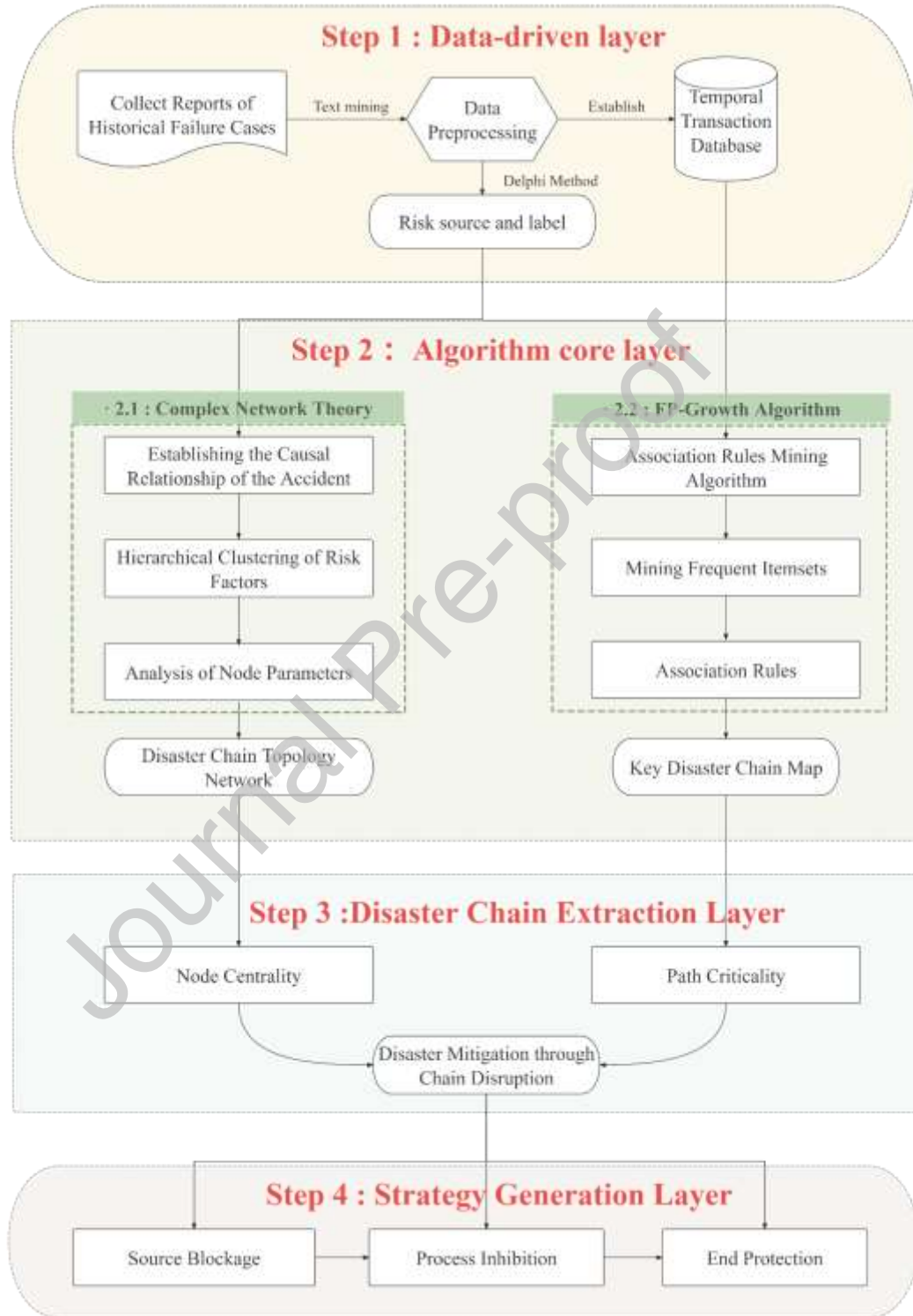


Fig. 2 Risk Mining Model Framework

This framework can achieve collaborative drive between complex network global topology

analysis and local causal mining of FP-Growth, breaking through the limitations of traditional methods in processing sparse time-series data and identifying implicit high risk rules. Relying on the Risk Intensity quantification model, it reshapes the paradigm of disaster-chain analysis, shifting from statistical correlation to causal inference and risk transmission.

3.2 Complex Network Theory

Complex network theory provides a powerful framework for modeling and analyzing systems composed of numerous interconnected elements. Its essence lies in abstracting the fundamental components of a complex system as nodes and their mutual interactions or relationships as edges, thereby constructing a network structure model (Feng, Y., Liu, Z., Jiang, Z., Xia, G., Cao, Y., Wang, X., Wang, H., 2023. Analysis of factors affecting ship collisions based on association rule mining and complex network theory. *Journal of Dalian Maritime University*. 49(3), 31-44. Wang, C., Ma, J., Su, C., Deng, J., Chen, W., 2025. Evolution and risk analysis of flood-induced coal mine disaster chains. *Journal of Safety and Environment*. Online first publication, 1-12.). Within the context of disaster chain risk research, hazard events, risk factors, and their evolutionary relationships in hydrogen pipelines can be effectively mapped onto complex networks.

Accordingly, we define a directed weighted complex network $G = \{V, E, W\}$ whose vertex set $V = \{V_1, V_2, \dots, V_n\}$ enumerates every event or state in the disaster chain, with n denoting the total number of nodes; the edge set E consists of directed links $V_i \rightarrow V_j$ that represent the causal or temporal progression whereby the occurrence or presence of V_i precipitates V_j ; and W is the weighted adjacency matrix whose entries W_{ij} quantify the probability, intensity, or other salient attributes of risk transmission along $V_i \rightarrow V_j$ (Li, T., Li, L., Mao, X., Sun, L., Wei, Y., 2025. Convenience evaluation method of bus stops based on complex network theory. *Journal of Chang'an University (Natural Science Edition)*. 45(3), 141-151. .

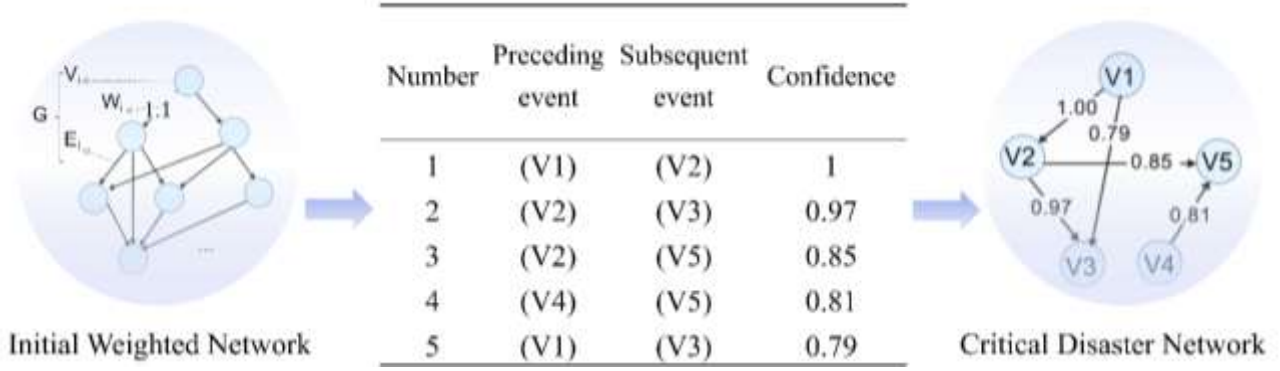


Fig. 3 Directed weighted network mapping schematic diagram

We employ a set of metrics spanning microscopic nodal distributions to macroscopic network averages to elucidate the network's topology and its role in risk contagion, evaluating node importance via degree, betweenness, and closeness centrality.

The node degree of V_i , denoted k_i , is defined as the total number of edges connected to it. In directed networks, this measure differentiates between in-degree and out-degree. The in-degree of V_i , denoted k_i^{in} , represents the number of edges pointing towards V_i and reflects its potential to be triggered as a consequence by preceding events within the disaster chain. Conversely, the out-degree of V_i , denoted k_i^{out} , equals the number of edges emanating from V_i and characterizes its capacity to act as an inducing factor triggering subsequent events. The expressions are:

$$k_i^{in} = \sum_{j=1, j \neq i} A_{ji} \quad (3.1)$$

$$k_i^{out} = \sum_{j=1, j \neq i} A_{ij} \quad (3.2)$$

where A_{ij} is the adjacency matrix ($A_{ij} = 1$ if edge (V_i, V_j) exists; $A_{ij} = 0$ otherwise).

Closeness Centrality for node V_i , denoted C_i , reflects the reciprocal of the average shortest path length from V_i to all other nodes in the network. A higher value indicates that V_i is closer to the network center, implying potentially faster propagation of information or risk throughout the network. The expression is:

$$CC_i = \frac{n-1}{\sum_{j=1, j \neq i} d_{ij}} \quad (3.3)$$

where d_{ij} represents the shortest path length between node V_i and node V_j .

Betweenness Centrality for node V_i , denoted BC_i measures the frequency with which V_i lies on

the shortest paths between all pairs of distinct nodes (V_j, V_k) in the network. A higher value signifies that V_i plays a more critical role in connecting different network segments and controlling risk flow, acting as a potential bottleneck or critical control point.

$$BC_i = \sum_{i \neq j \neq k} \frac{\sigma_{jk(i)}}{\sigma_{jk}} \quad (3.4)$$

where σ_{jk} is the total number of shortest paths between nodes V_j and V_k , and $\sigma_{jk(i)}$ is the number of those shortest paths that pass through node V_i .

3.3 modified FP-Growth algorithm

In the context of association rule mining, let $I = \{i_1, i_2, \dots, i_m\}$ denote the universal set of items (Ma, X., Zhang, R., Qiao, W., Han, B., Yang, J., 2025. Analysis of the Seafarers' Unsafe Actions Causality Network Based on Association Rule Mining. Journal of Transport Information and Safety. Online first publication, 1-10.

. An itemset X is a subset of I , and a transaction T is a subset of items (typically representing a record or an event). Association rules are expressed in the form $X \rightarrow Y$, where X and Y are disjoint itemsets. Traditional association rules rely on support, confidence, and lift to measure the strength of rules. However, these indicators only consider the frequency of events and ignore the inherent differences in the severity of risk events. Low-frequency, high-harm events may be more destructive than high-frequency, low-risk events. Therefore, we proposed a risk-weighting mechanism, the core of which lies in the adoption of two new indicators: Risk Weighted Support (RWS) and Risk Weighted Confidence (RWC), which are the key criteria for us to determine the critical rule chains. Currently, there is no single, independent national standard for the risk assessment of hydrogen pipelines. The weight assignment of disaster points mainly refers to the event consequence levels in the hydrogen infrastructure failure database HyRAM, and the relative weights of each risk item are determined by combining the analytic hierarchy process.

$$RWS(I) = \frac{count(I)}{N} \times \left[\sum_{T \subseteq I} \left(\frac{\sum_{i \in I} w_i}{count(I) \times |I|} \right) \right] \quad (3.5)$$

where N is the total number of transactions, and w_i is the severity weight of risk item I .

In the modeling of causal event chains, to avoid redundancy caused by long chains and to

highlight the key causal logic, a "time attenuation factor" λ can be introduced to limit the maximum allowable interval between adjacent causal events. This ensures that only short-term causal chains with high confidence are retained, eliminating redundancy while accurately reflecting the core characteristics of hydrogen disasters, which are "instantaneous triggering - rapid evolution".

$$RWC(X \rightarrow Y) = \frac{RWS(X \cup Y)}{RWS(X)} \times e^{-\lambda \cdot |\Delta t_{X \rightarrow Y}|} \quad (3.6)$$

where λ represents the time decay coefficient. The smaller the λ value, the stronger the penalty for long intervals. Based on the temporal characteristics of hydrogen energy accident chains, three typical evolution paths of hydrogen pipeline failure accidents and the corresponding attenuation factor values have been listed : When hydrogen leaks and causes hydrogen combustion (Patel, P., Garaniya, V., Baalisampang, T., Arzaghi, E., Abbassi, R., Salehi, F., 2024. Enhancing safety in hydrogen refuelling stations: Computational analysis of hydrogen explosion hazard. The Canadian Journal of Chemical Engineering. 103(1), 82-97.), if an ignition source is present, the fire will be triggered instantaneously within 0 - 5 seconds. In this case, the solution is as follows : $\lambda \approx 0.871$. If the hydrogen leakage does not cause hydrogen combustion before an explosion occurs, the time window during which hydrogen diffuses to the explosive concentration range is 0-100 seconds. Solving this gives : $\lambda \approx 0.993$. If a fire occurs first and then leads to an explosion, the critical time for the explosion to occur due to local high temperature is usually 10-60 seconds, which is 0-100 seconds. Solving this gives : $\lambda \approx 0.989$.

Based on Eq (3.7), key risk rules are extracted. This not only reduces data redundancy but also highlights the causal paths that truly have engineering risk significance.

$$R_{risk} = \{X \rightarrow Y \mid RWC(X \rightarrow Y) \geq \theta_{risk} \cap RWS(X \cup Y) \geq \sigma_{risk}\} \quad (3.7)$$

where θ_{risk} is the risk confidence threshold value and σ_{risk} is the risk support threshold value.

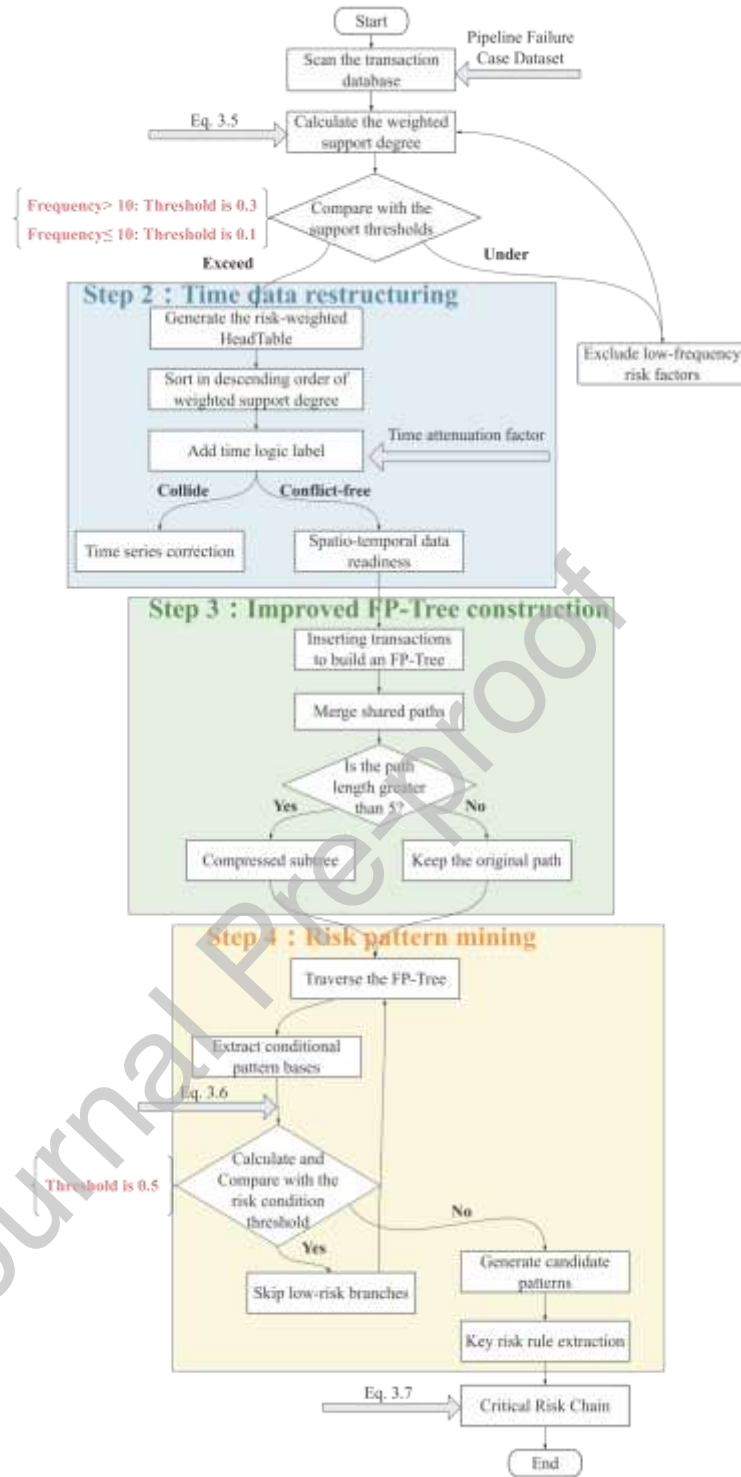


Fig. 4 The complete process of mining risks using the modified FP-Growth algorithm

As shown in Fig. 4, the mining process of the modified FP-Growth algorithm is introduced. First, scan and calculate the weighted support of each risk factor in the transaction database, generate a header table sorted in descending order of weighted support, and eliminate infrequent items. Next, reorder the items in each transaction according to the header table and add time tags to remove item sets that violate the time-order logic. Then insert the re-organized transactions into the FP-tree, merge paths

with the same time prefix, compress sub-trees whose path length exceeds 5, and skip low-support branches when generating conditional pattern bases. Finally, traverse the FP-tree, prune the node chains lacking specific data items, output the final frequent item sets, and visually display the key risk chains.

4. Case Analysis of Key Risk Chain Identification for Hydrogen Pipeline

4.1 Collection of hydrogen pipeline accident case data

The source data for this study was derived from publicly available incident reports related to hydrogen accidents, documented by the Joint Research Centre of the European Commission up to March 2024 (Wen, J. X., Marono, M., Moretto, P., Reinecke, E. A., Sathiah, P., Studer, E., Vyazmina, E., Melideo, D., 2022. Statistics, lessons learned and recommendations from analysis of HIAD 2.0 database. International journal of hydrogen energy. 47(38), 17082-17096.. We define our scope as hydrogen pipelines, explicitly excluding equipment failures in storage areas or power plants to ensure research consistency. We then reviewed and cleaned the data based on the completeness and availability of accident reports. Early records with unclear causes or consequences were removed. After this screening, fewer than 30 cases from the EU database met the above criteria. To avoid geographical or typological bias, we systematically supplemented the dataset with additional publicly reported incidents from verified news sources and peer-reviewed literature. This step ensured coverage of multiple regions and countries, as well as diverse accident types. Through this systematic screening process, the final set of 50 accident cases forms a high-quality, severe, and representative sample for analysis.

Table 1 Typical hydrogen pipeline accident cases at home and abroad

Time	Location	Accident	Time	Location	Accident	Time	Location	Accident
2002 - 01	Japan	Fire	2009 - 04	France	Leak no ignition	1979 - 02	US	Explosion
2007 - 06	France	Explosion	2019 - 06	Norway	Explosion	1985 - 07	Saudi Arabia	Fire
Time	Location	Accident	Time	Location	Accident	Time	Location	Accident

2005 - 09	France	Explosion	2019 - 03	Belgium	Fire	1985 - 12	US	Explosion
1995 - 1	Germany	Explosion	2013 - 04	France	Leak no ignition	1991 - 1	Japan	Fire
2007 - 09	Ukraine	Explosion	2010 - 02	France	Leak no ignition	1974 - 11	US	Explosion
2004 - 01	France	Leak no ignition	2011 - 05	US	Explosion	1974 - 07	UK	Leak no ignition
2000 - 06	US	Fire	2021 - 03	Belgium	Explosion	1985 - 05	Japan	Explosion
1998 - 09	Germany	Explosion	2020 - 01	China	Explosion	1985 - 02	Japan	Fire
1997 - 07	Germany	Explosion	2015 - 03	China	Explosion	1997 - 01	Europe	Explosion
1991 - 02	South Korea	Explosion	2021 - 07	China	Explosion	2007 - 1	Netherlands	Fire
2001 - 09	Finland	Explosion	2003 - 03	China	Explosion	2001 - 1	France	Fire
2002 - 1	Belgium	Explosion	2021 - 02	US	Explosion	2019 - 05	Belgium	Leak no ignition
2017 - 12	Italy	near miss	2022 - 03	US	Explosion	2019 - 05	US	fire
2018 - 11	US	fire	2001 - 2	China	fire	2020 - 7	China	Explosion
2015 - 04	US	near miss	2011 - 4	China	leak no ignition	2002 - 4	America	Explosion
2021 - 11	Spain	leak no ignition	2025 - 2	China	Explosion	2023 - 1	China	fire

2025 - 3	Korea	leak no ignition	2019 - 6	America	Explosion
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The statistical analysis of hydrogen pipeline failure incidents reveals that over 70% of the cases triggered secondary accidents, indicating a pronounced propensity for cascading risks in such events. Notably, approximately 50% of the incidents involved pipeline explosions, a proportion that significantly surpasses other accident consequences, underscoring the severity and high incidence of explosive events in hydrogen pipeline operations. Geographically, the failure cases were predominantly concentrated in Europe and Asia, suggesting these regions may represent higher-risk zones for hydrogen pipeline infrastructure. These findings highlight the critical need for enhanced safety protocols, robust design standards, and rigorous maintenance practices to mitigate explosion risks and prevent secondary accidents in hydrogen pipeline systems.

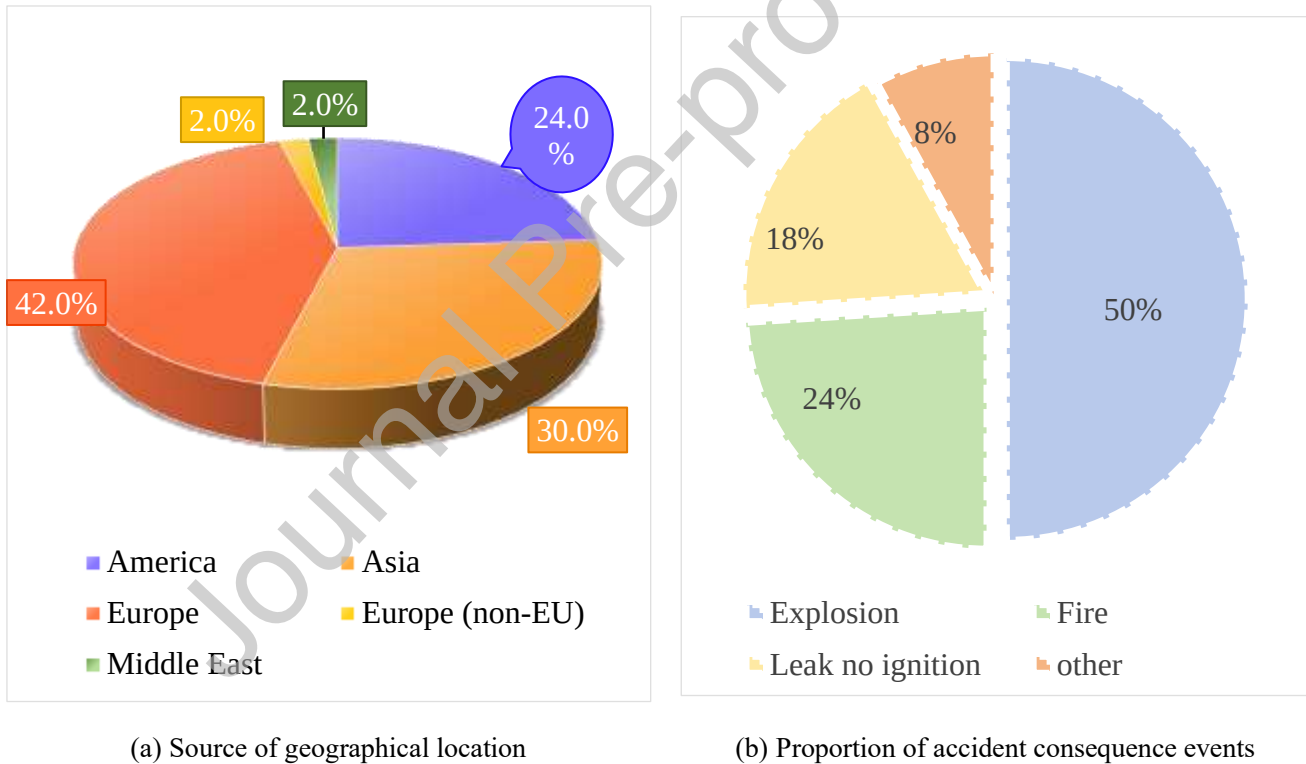


Fig. 5 Statistics on hydrogen pipeline failure cases

4.2 Risk source system construction and structured annotation

Grounded in disaster chain theory, a dedicated text corpus was constructed using the accident case descriptions presented in Table 1. Structured textual analysis of the accident causation narratives was then performed utilizing Python programming in conjunction with the Jieba library for Chinese text segmentation. This methodology enabled the systematic development of a specialized corpus detailing

the causes of accidents in hydrogen pipeline systems and related facilities. Subsequently, to ensure systematic and authoritative identification of risk factors, the Delphi method was applied for statistical analysis and refinement of the initially identified risk factor set. To guarantee the quality of the expert consultation, 15 specialists with over three years of experience in the pipeline industry were invited. Their expertise covered hydrogen safety, pipeline integrity management, risk assessment, and accident analysis. The expert panel consisted mainly of university researchers and senior engineers from pipeline network companies. In the first round, the initial list of 52 risk factors derived from text mining was submitted to the experts. They were asked to evaluate the relevance of each factor and suggest additions or modifications. The feedback was collected and analyzed, leading to a revised list. In the second round, the experts received the summarized results and were asked to re-evaluate the items, particularly those showing significant divergence in the first round. Consensus was considered achieved when over 80% of the experts rated a risk factor as “highly relevant” or “relevant”. After being selected and confirmed through the Delphi method, this study classified the obtained risk nodes according to their main functional roles and the initial stage when they first emerged. The results are shown in Table 2. It aims to establish a modeling framework based on the dominant roles for the subsequent quantitative analysis model, thereby clearly defining the causal transmission relationships between the nodes.

Table 2 Classification of Risk Nodes and Main Functional Roles of Hydrogen Pipeline Disaster Chains

Main stages and functional roles	Chains					
	Label	Risk Source	Label	Risk Source	Label	Risk Source
Gestation Stage (Causes and Precursors)	V1	Corrosion	V9	Welding Defect	V17	Inadequate Support
	V2	Material Defect	V10	Loose Flange	V18	Condensate Erosion
	V3	Seal Failure	V11	Bolt Failure	V19	Ice Blockage
	V4	Operational Error	V12	Gasket Fatigue	V20	Hydrogen Attack

	V5	Failure to Inert	V13	External Impact	V21	Sulfur Corrosion
	V6	Temperature Change	V14	Excavation Work	V22	Friction Spark
Continued Table						
	V7	Pressure Fluctuation	V15	Ground Subsidence	V23	Electrostatic Ignition
	V8	Design Defect	V16	Vibration Stress		
Disaster Occurrence Stage (Triggering and Upgrading)	V24	Equipment Overheating	V28	Fire	V32	Secondary Fire
	V25	Gas Mixture	V29	Explosion	V33	Pipeline Rupture
	V26	Leakage	V30	Detonation		
	V27	Accumulation	V31	Jet Fire		
Accident Impact Stage (The ultimate consequence)	V34	Equipment Damage	V38	Plant Shutdown	V42	Asphyxiation due to Oxygen Deficiency
	V35	Building Destruction	V39	Traffic Disruption	V43	Secondary Pollution
	V36	Casualties	V40	Resident Evacuation	V44	Loss of Corporate Credibility
	V37	Minor Injuries	V41	Economic Loss	V45	Public Panic

In Fig. 6, we comprehensively visualize the risk nodes and their causal relationships across the three canonical phases of disaster evolution. These phases are the pre-disaster incubation, the disaster-triggering, and the impact-propagation. Using internationally recognized failure taxonomies from authorities such as EGIG and PHMSA, we aggregate the 23 pre-disaster risk nodes from Table 2 into five archetypal hazard clusters. These clusters are Corrosion, Damage by third parties, Geological disasters, Inherent defects, and Misoperation. Transitioning to the disaster-triggering stage, these risk nodes are further regrouped according to their dominant physical failure mechanisms into three

representative consequence regimes. In the subsequent impact-propagation stage, and according to an overarching disaster-chain systems-theoretic assessment framework, the twelve nodes listed in Table 2 are grouped into four receptor dimensions. These dimensions are personnel safety, economic loss, societal disruption, and environmental degradation. This multi-layered, multi-dimensional taxonomy of risk nodes not only clarifies the subordinate relationships among risk factors but also furnishes the theoretical substrate for the subsequent construction of a dynamic gradient weighting system.

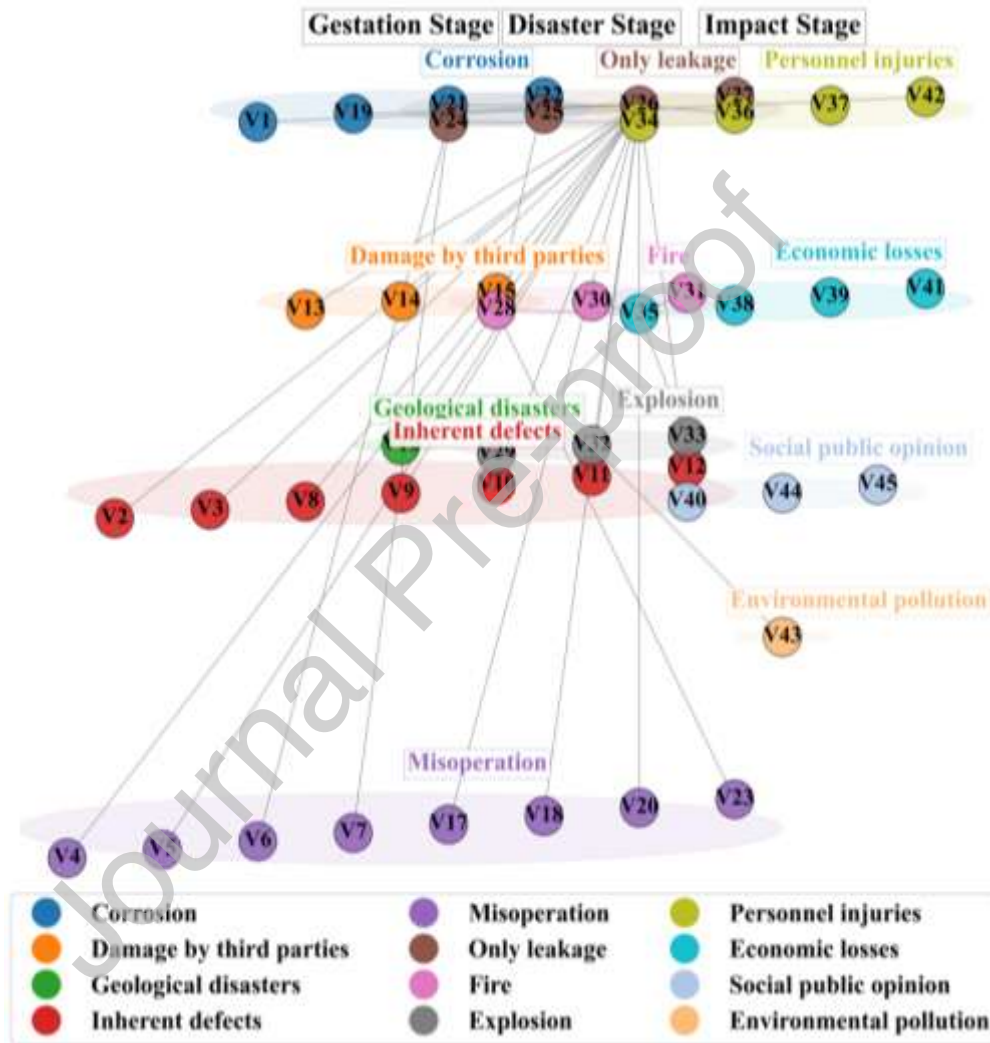


Fig. 6 Hierarchical Clustering of Hydrogen Pipeline Risk Factors

4.3 Analysis of Key Nodes in Disaster Chains Based on Complex Networks

In Fig. 7, we constructed an initial network topology model of hydrogen pipeline disaster chain based on complex network theory. By abstracting the key risk factors and their interaction relationships in the disaster evolution process, the multi-level propagation paths of the disaster chain were visually characterized. The network's scale-free structure is marked by key hubs such as V26, V28, and V29,

which have node degrees 2 – 3 times the network average, designating them as critical for disaster propagation.

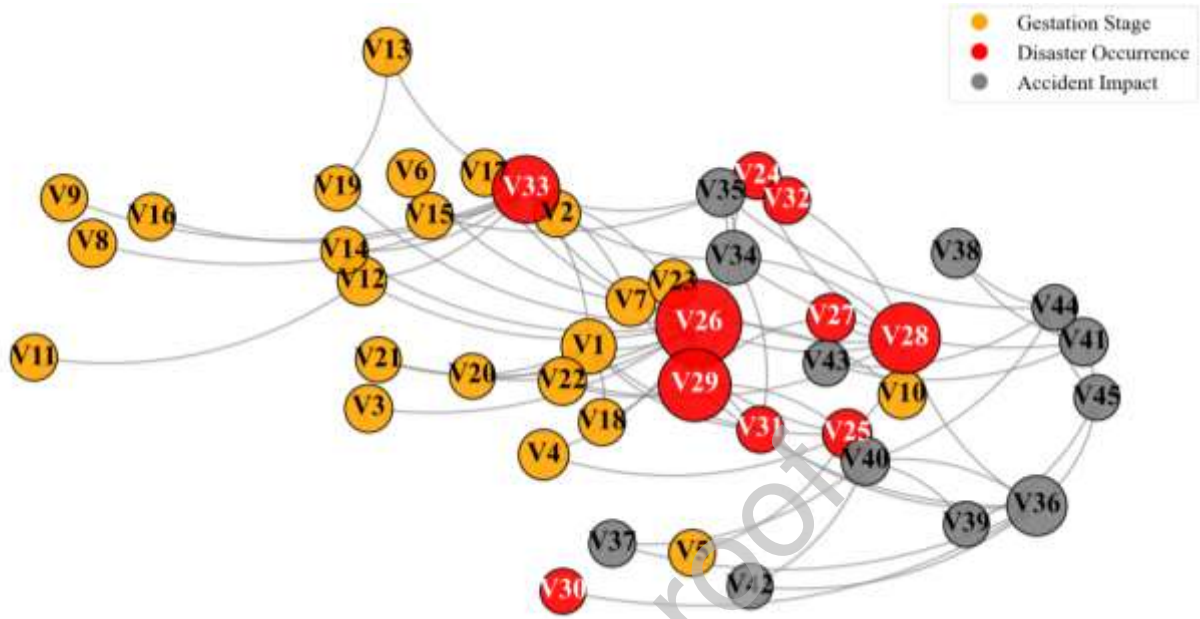


Fig. 7 Initial network topology diagram of the disaster chain

In Fig. 8, we further quantify the controlling influence of critical nodes within the disaster chain through comprehensive topological metrics including node in-degree/out-degree, closeness centrality, and betweenness centrality. Analytical results demonstrate that corrosion exhibits high out-degree during the gestation stage, indicating that initial management or technical deficiencies can significantly amplify subsequent risks through systemic vulnerabilities. Given the inherent risk of rapid escalation in hydrogen accidents, leakage and explosion nodes show high closeness centrality. A leak can form an explosive mixture in seconds, triggering a catastrophic chain of events that endangers both people and the environment. In contrast, gestation nodes such as welding defects display lower closeness centrality, suggesting progressive risk evolution that may culminate in abrupt failures through long-term accumulation. The elevated betweenness centrality of pipeline rupture underscores the paramount importance of pipeline integrity management, as its failure can trigger secondary disasters including catastrophic leakage or explosion. Consequently, we propose an early diagnostic strategy based on real-time pressure fluctuation spectrum analysis. This approach involves deploying high-frequency sensors at critical nodes, establishing dynamic warning thresholds, and enhancing system responsiveness. Ultimately, the collected fluctuation data can be integrated into a decision support system to drive data-driven predictive maintenance planning.

A composite importance score for the top 10 critical nodes was derived through the integrated quantification of in-degree, out-degree, closeness centrality, and betweenness centrality metrics, enabling holistic assessment of nodal influence across multiple topological dimensions.

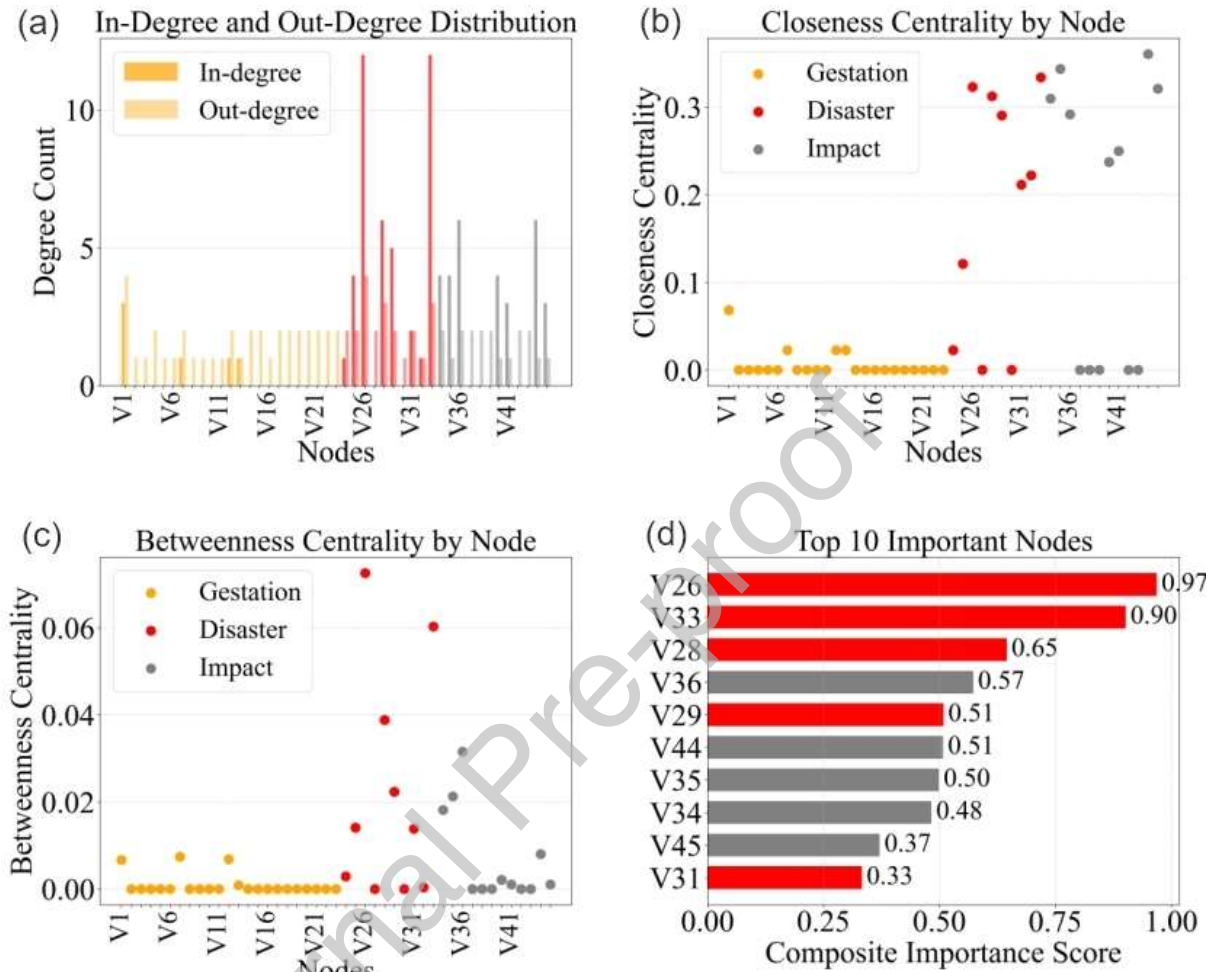


Fig. 8 Analysis of Network Node Topology Indicators

4.4 High-risk disaster chain identification based on the modified FP-Growth algorithm

4.4.1 Risk source coupling relationship

In the realm of pipeline system risk investigation, the inherent nature of risk sources to remain dormant over extended periods and their propensity to conceal their interconnections render it arduous to directly and comprehensively discern their mutual associations. To circumvent this challenge, the FP-growth algorithm was employed to conduct a meticulous localized analysis of 23 salient risk sources. As depicted in Fig. 9 and Fig. 10, the extracted high-confidence risk source chain offers a lucid revelation: intrinsic pipeline defects such as gasket failure and sealant failure are highly susceptible to coupling with external ignition factors like friction sparks. This interaction can precipitate a cascade of severe secondary disasters, including leaks and fires, thereby posing a

significant threat to the safe and stable operation of pipeline systems. Delving deeper into the etiology of these failures, third-party excavation activities emerge as the most probable cause of the aforementioned failure phenomena.

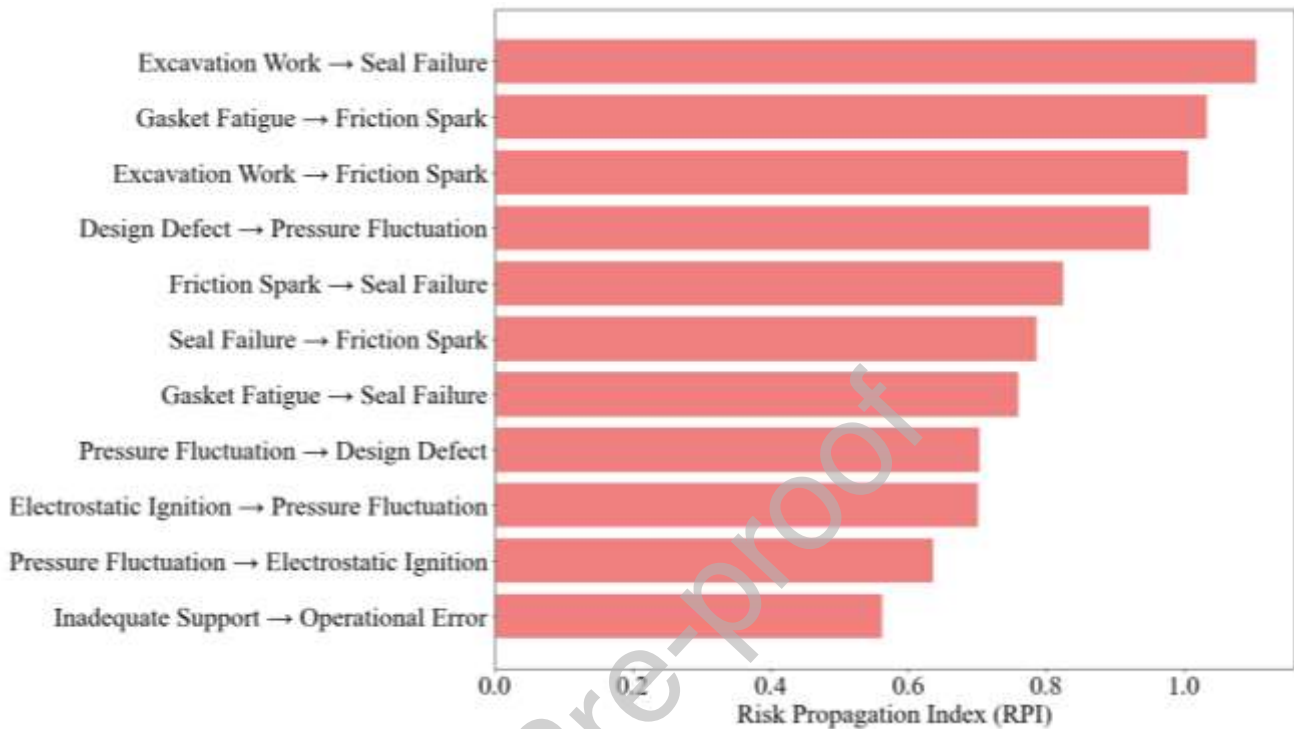


Fig. 9 Risk transmission index among risk sources

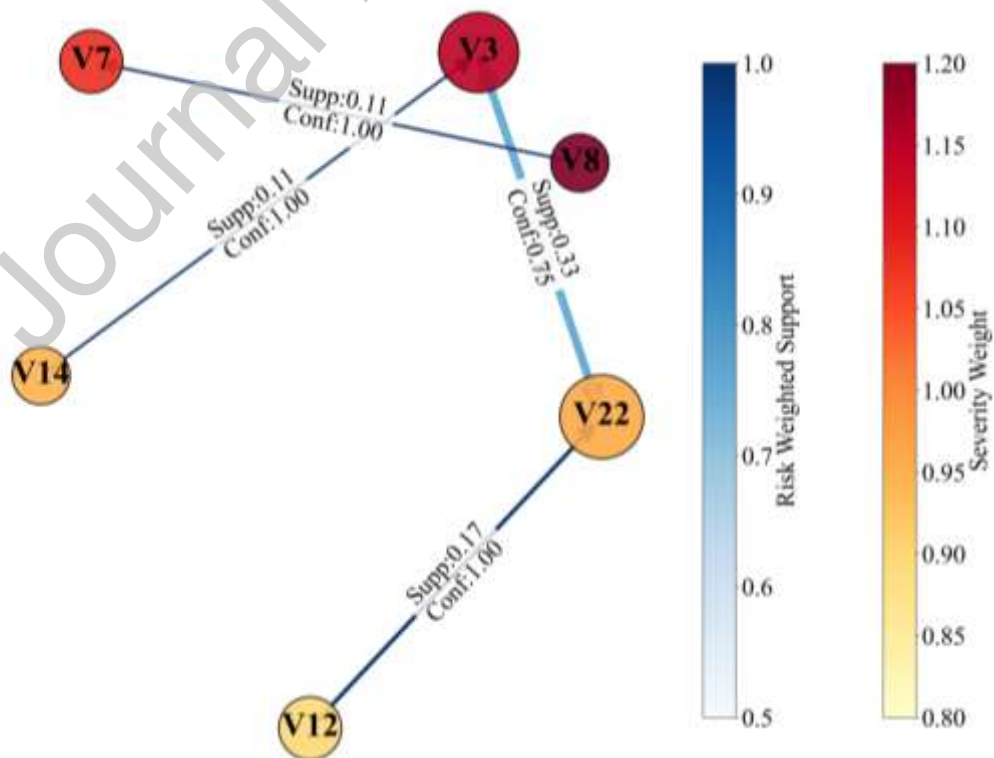


Fig. 10 Risk source association rule graph

4.4.2 Impact consequence association rule

This study employed a modified FP-growth algorithm. First, the weighted support values for all item categories were computed. Itemsets with weighted support values below the 0.3 threshold were deemed insufficiently supported and were consequently excluded from FP-tree construction, as depicted in Fig. 11. The remaining single frequent itemset was then placed into the item header table and sorted in descending order of support.



Fig. 11 Pre-mining screening of risk nodes based on weighted support

Subsequently, the risk weighted confidence (RWC) was calculated for each frequent itemset, leading to the extraction of a high-reliability rule set, as presented in Table 3. For instance, the rule $\{V26 \rightarrow V29 \rightarrow V36\}$ (leakage $\rightarrow$ explosion $\rightarrow$ casualties) exhibited an RWC value as high as 0.896. This value clearly indicates that this particular pathway has a highly significant amplification effect in the propagation of disasters, implying that once a leakage occurs, the probability of it subsequently triggering an explosion and leading to casualties is extremely high. To provide a more intuitive visualization of the distribution characteristics of strong association rules, a three-dimensional scatter plot was utilized in Fig. 12. Furthermore, in conjunction with Table 3, it was found that rules with high confidences and high risk-weighted lifts are predominantly concentrated in pathways related to explosions and leakage.

Table 3 Strong association rules and confidence degree

Strong association rule	RWS	RWC	Strong association rule	RWS	RWC
$\{V26 \rightarrow V33 \rightarrow V28\}$	0.336	0.923	$\{V28 \rightarrow V26\}$	0.486	0.827

$\{V26 \rightarrow V36 \rightarrow V29\}$	0.401	0.896	$\{V29 \rightarrow V26\}$	0.588	0.784
$\{V36 \rightarrow V29\}$	0.480	0.889	$\{V33 \rightarrow V26\}$	0.364	0.758
$\{V35 \rightarrow V26\}$	0.390	0.882	$\{V36 \rightarrow V26 \rightarrow V29\}$	0.401	0.743
$\{V33 \rightarrow V28\}$	0.406	0.846	$\{V33 \rightarrow V26 \rightarrow V28\}$	0.336	0.700
$\{V29 \rightarrow V36 \rightarrow V26\}$	0.401	0.836	$\{V26 \rightarrow V28 \rightarrow V33\}$	0.336	0.691
$\{V36 \rightarrow V26\}$	0.448	0.830	$\{V28 \rightarrow V33\}$	0.406	0.690
$\{V28 \rightarrow V33 \rightarrow V26\}$	0.336	0.828	$\{V26 \rightarrow V29 \rightarrow V36\}$	0.401	0.683

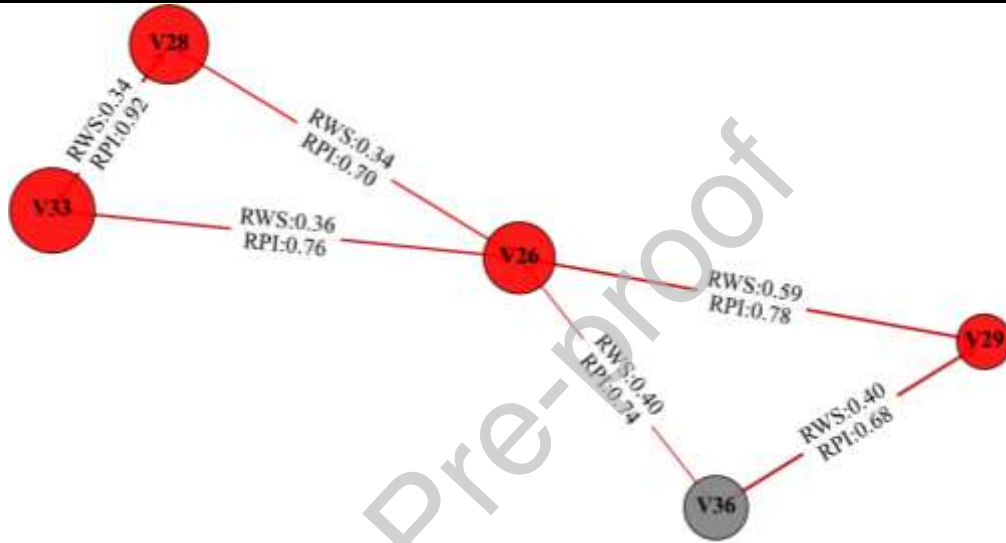


Fig. 12 Impact consequence association rule graph

5. Optimization of disaster chain Prevention Strategies for Hydrogen Pipeline Disasters

In the progression of hydrogen accidents, leakage and explosion are critical stages that often serve as the core factors for the rapid escalation and propagation of accidents. Once issues arise in these stages, the severity of the accident can escalate dramatically within a short period, subsequently triggering a series of more severe consequences, such as casualties. Based on association rule mining and risk transmission path analysis, this study identifies two high-risk disaster chains in hydrogen transportation systems. Gasket fatigue and sealant failure are key factors leading to leakage, primarily resulting from the combined effects of long-term mechanical stress, material aging, and chemical corrosion in the operating environment. To address these risk formation mechanisms, a three-tier prevention and optimization framework is proposed as shown as Fig.13.

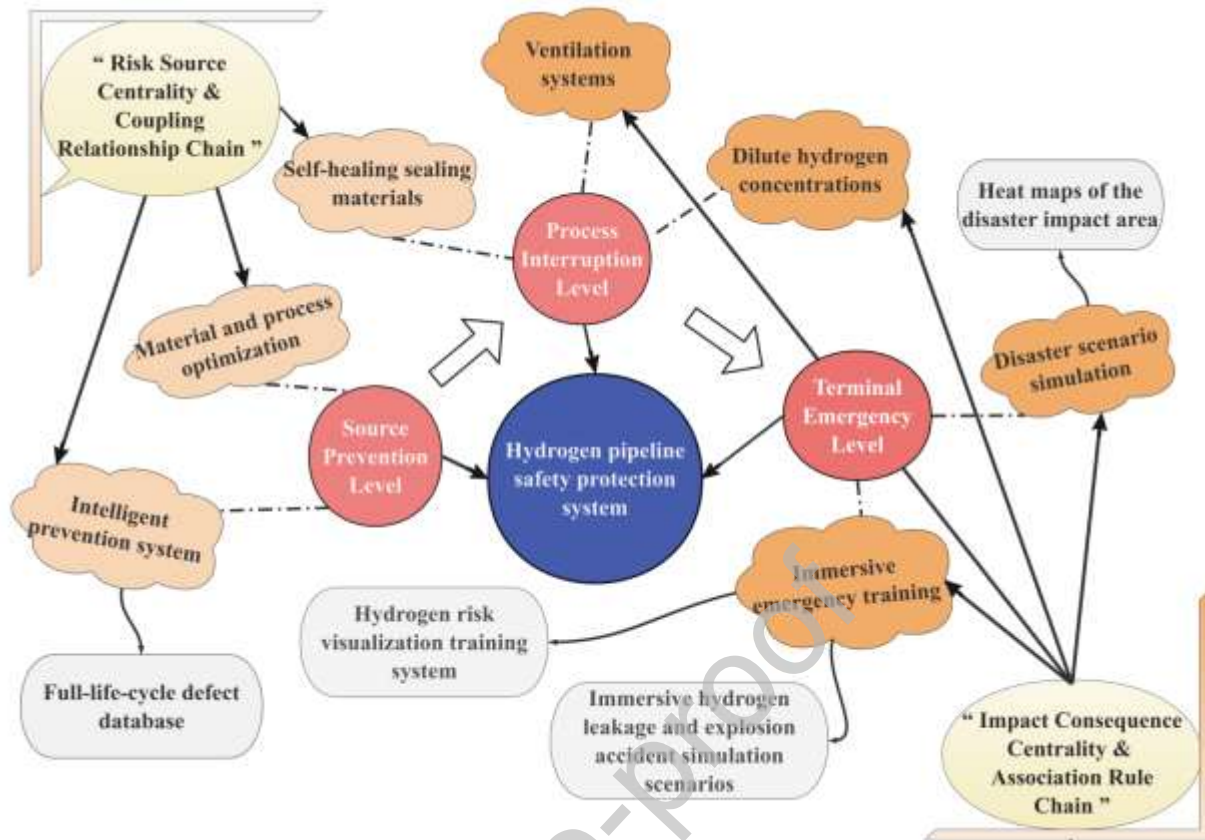


Fig. 13 Three-tier prevention and optimization framework

At the source prevention level, it is recommended to establish an intelligent prevention system based on a full-life-cycle defect database. This involves developing a full-life-cycle defect monitoring and early warning mechanism, which monitors the operational status of key components in hydrogen transportation systems in real time. By integrating historical data and failure mode analysis, this mechanism can identify potential defects and hazards in advance, providing a scientific basis for preventive maintenance. Additionally, material and process optimization are essential. Based on in-depth analysis of the failure causes of key components such as gaskets and sealants, more durable and corrosion-resistant materials should be selected, and manufacturing processes should be optimized to reduce leakage risks at the source.

At the process interruption level, dynamic monitoring and intelligent interruption capabilities must be enhanced. The implementation of advanced dynamic sealing technologies, such as self-healing materials or intelligent systems, is critical to prevent leaks. Any escaping hydrogen must be diverted to safe areas and promptly diluted by ventilation systems to prevent combustible concentrations from forming.

At the terminal emergency level, a comprehensive risk mitigation strategy is necessary. A three-

dimensional prediction model for hydrogen diffusion should be established, integrating fire dynamics theory with the actual parameters of hydrogen transportation systems. This model dynamically generates heat maps of the disaster impact area, providing emergency responders with intuitive risk distribution information to facilitate precise rescue and risk mitigation. Furthermore, immersive hydrogen leakage and explosion accident simulation scenarios should be constructed using virtual reality technology. A hydrogen risk visualization training system should be developed to enhance specialized emergency skills and improve the safety awareness and response capabilities of personnel, especially those in high-consequence areas with dense pipeline networks.

6. Conclusions

In summary, based on the disaster chain system theory, this study has constructed a four-dimensional analysis framework for hydrogen pipeline networks, encompassing the disaster environment, hazardous factors, disaster-bearing carriers, and disaster consequences. This framework systematically dissects the transmission mechanisms underlying the evolution of disasters from the incubation of risks to their full manifestation. By integrating data from the European Hydrogen Incidents and Accidents Database with other public records, and leveraging complex network theory, this study innovated the FP-Growth algorithm's itemset weighted evaluation module. This integrated methodology enabled the precise mining of high-risk sources and frequent itemsets associated with severe disaster outcomes. Through the calculation of weighted risk support and risk-weighted confidence, two high-risk disaster chains driven by intrinsic defects and third-party damage were identified. Furthermore, recommendations for optimizing disaster chain prevention and mitigation measures were proposed, integrating strategies of source control, process interruption, and terminal protection. Future research should progressively expand the scope of investigation to recognize intersecting influence pathways, thereby facilitating cross-chain collaborative management.

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Declaration of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.